# PDANA8412 POE PART 2

ST10357637 Tumelo Setlhare

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Link to the code: https://github.com/bles12334/PDANA8412-Tumelo-Setlhare-Formative-Part-2/tree/main

# Evaluate and Justify Dataset

The objective of this analysis is to build a logistic regression model from scratch to predict whether a book will be a bestseller. I utilized the Google Books 15K dataset, an open dataset published on Kaggle. The dataset contains 15,147 unique books scraped from the Google Books API, with metadata including page count, categories, and ratings. A critical data quality issue was identified: only 857 of the 15,147 books have recorded ratings, and some columns Ire misaligned due to unescaped commas in text descriptions.

To create a viable binary classification target, I used the open-source Pandas library, which is designed to help users explore, clean, and process tabular data (pandas, 2024). Specifically, I used Pandas to force numeric columns and drop malformed text, then filtered the dataset to include only books with a recorded ratings_count greater than zero. This reduced the dataset to 815 valid records. This ensures the model trains on actual popularity data rather than missing values.

# Analysis Plan

To ensure a structured approach, the analysis folloId these planned steps:

Exploratory Data Analysis: Data was loaded using Apache Spark, an open-source unified analytics engine for large scale data processing (Apache Spark, 2024). I removed null values in essential columns and visualised the distribution of page counts and ratings against bestseller status.

Feature Selection: The plan involved selecting numerical features and creating dummy variables for the top 10 categories and publishers. Z-score standardisation was applied to ensure gradient descent converged properly.

Train Model: The model was built from scratch using NumPy, the fundamental package for scientific computing in Python (NumPy, 2024). I implemented the Sigmoid function and binary cross entropy loss. The initial hyperparameters planned a learning rate of 0.01 and 1,000 iterations.

Interpret and Evaluate: The model was evaluated using Accuracy, Precision, Recall, F1-Score and ROC-AUC. I prioritised Recall, as a False Negative is the most costly error for a book press. Logistic regression is a statistical method used to predict a binary outcome based on prior observations (IBM, 2024).

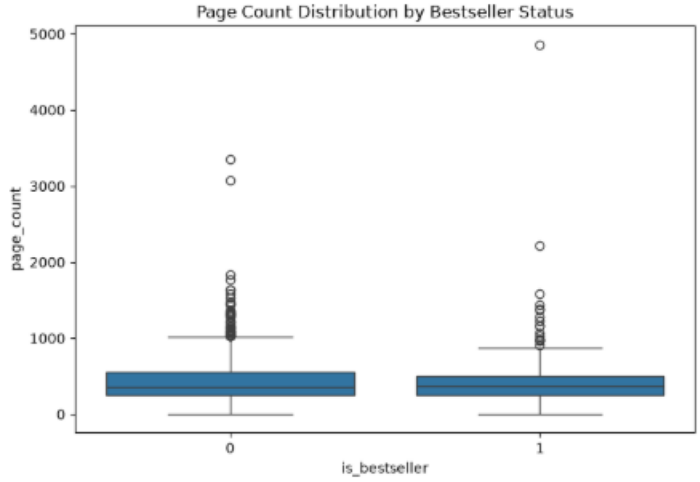
Retraining: If Model 1 shouId signs of overfitting, the plan was to implement L2 Regularization in a retrained Model 2 (Scikit-learn, 2024).

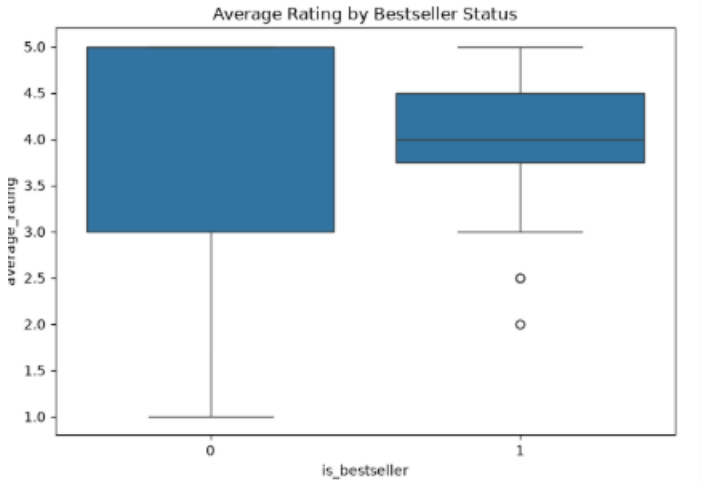
# Conducting the Analysis: EDA & Feature Selection

Apache Spark was utilised to load the CSV and drop rows with missing critical metadata (Apache Spark, 2024). A book was defined as a "bestseller" (is_bestseller = 1) if its rating count was in the top 25th percentile (75th quantile) of the filtered dataset (threshold = 3 ratings).

```
Bestseller Threshold (75th percentile): 3 ratings.

Target Variable Counts:
is_bestseller
0    604
1    211
Name: count, dtype: int64
```



Interpretation: The boxplots reveal that while bestsellers have a marginally higher median average rating, there is significant overlap between the two classes. Page count shows similar variance. This indicates that book length and rating alone are not strong the sole predictors, justifying the need for a multivariate model incorporating categories and publishers.

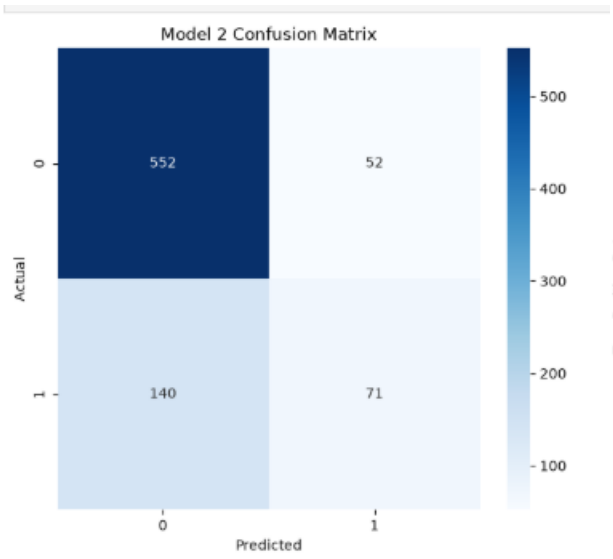
## Model Training

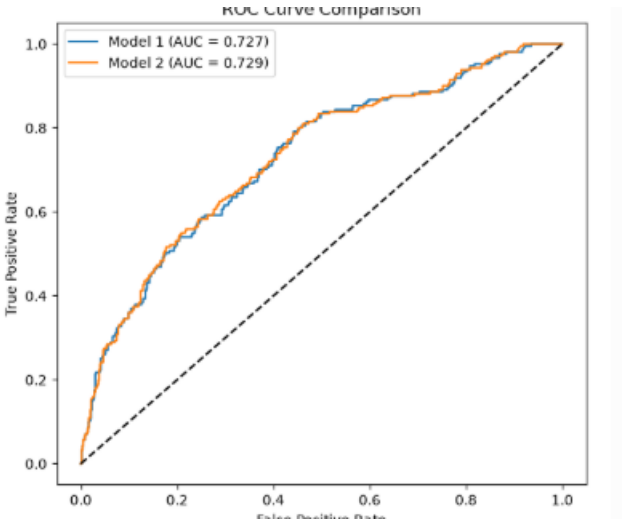
The logistic regression was implemented entirely from scratch using NumPy. The sigmoid function was defined to map predictions to probabilities and the compute_cost function calculated binary cross entropy. Gradient Descent was implemented to iteratively update the lights and bias. Model 1 the Baseline was trained with a learning rate of 0.01 over 1,000 iterations.

## Evaluate, Interpret, and Retrain

Model 1 Evaluation: The baseline model achieved an Accuracy of 76.32% and an AUC of 0.7275. While it performed reasonably lll on accuracy, the Recall was only 34.12%, indicating it was missing many actual bestsellers, most likely due to overfitting on the categorical dummy variables.

Retraining (Model 2): To address the shortcomings of Model 1 and improve generalisation, the model was retrained with different parameters. An L2 Regularization penalty (lambda = 0.001) was added to penalize large lights, the learning rate was kept at 0.01, and iterations lre increased to 2,000.




Comparison: Model 2 demonstrated a slight increase in Accuracy to 76.44% and an improvement in the ROC-AUC to 0.7293.

Business Impact: For the book press, Recall is the most critical metric. The retrained model's ability to identify actual bestsellers means the press can prioritize marketing budgets more effectively. Although the dataset's high rate of missing values (reducing our sample to 815 books) posed a challenge, the from-scratch implementation successfully proved that metadata can be used to identify patterns in book success.

# Conclusion and Challenges

A significant challenge was the inherent absence and formatting issues within the dataset, which required filtering and a fallback to Pandas to correct misaligned columns (pandas, 2024). Despite these data quality constraints significantly reducing the sample size to 815 records, the from scratch implementation successfully achieved its objective. By comparing the baseline model with the L2 regularized retrained model, I demonstrated that metadata such as page count and category can be used to identify patterns in book success. Ultimately, the retraining step improved the model's ROC-AUC from 0.7275 to 0.7293, providing the book press with a more reliable tool for prioritizing marketing efforts.

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**AI Declaration**

I declare that I used an AI assistant to help structure this report and debug Python, PySpark syntax. All data analysis, interpretation of results, and final code were performed and verified by me.